Report 3 Cleaning using a nextflow pipeline

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Transciptomics

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1 Clean pipeline

Additionally, we developed a **Nextflow** pipeline in order to make this preprocessing by the automation of the FastQC reports and also the fastp cleaning. This is made by extracting the features that the FastQC output has in the following way:

For each category: - Basic Statistics - Per base sequence quality - Per sequence quality scores - Per base sequence content - Per sequence GC content - Per base N content - Sequence Length Distribution - Sequence Duplication Levels - Overrepresented sequences - Adapter Content

While searching for `FAIL` in each category in order to assign the following fastp flags:

QC feature	fastp flag
Per base sequence content	–trim_front
Sequence Duplication Levels	–dedup
Overrepresented sequences	–overrepresentation_analysis

For that parsing, we are using the following basic bash script (awk as the main language).

```
#| eval: false
#review the Per base sequence content field
base_cont=$(grep "Per base sequence content" <(echo -e $sum) | cut -f1 | grep "FAIL" | uniq)
if [[ $base_cont=="FAIL" ]]; then
    base_cont="--trim_front1 ${trim_fron}"
else
    base_cont=""
fi

#review the Sequence Duplication Levels
dup_lev=$(grep "Sequence Duplication Levels" <(echo -e $sum) | cut -f1 | grep "FAIL" | uniq )
if [[ $dup_lev=="FAIL" ]]; then
    dup_lev="--dedup"
else
    dup_lev=""
fi

#review the over represetation sequences
over=$(grep "Sequence Duplication Levels" <(echo -e $sum) | cut -f1 | grep "FAIL" | uniq)
if [[ $over=="FAIL" ]]; then
    over="--overrepresentation_analysis"
else
    over=""
fi
```

This code generates the QC-dependent arguments that we use to clean the fastq files and make sure it works with both paired and unpaired data. > It can be reviewed in more detail in the `src/run_fastp.sh` script

1.1 Input

For usage, we have to provide a table without a header but with the SRR ID column and the paths for fastq data, so we are using the metadata table to construct it in a: `<srr_id>|<path1>|[path2]` format that we generate:

```
meta_clean = pd.read_csv("data/metadata_clean.csv", sep="\t")
# build the data frame
meta_cleaning = pd.DataFrame(meta_clean["srr_id"])
meta_cleaning["path_1"] = (
    "data/" + meta_cleaning["srr_id"] + "/" + meta_cleaning["srr_id"] + "_1.fastq"
)
meta_cleaning["path_2"] = (
    "data/" + meta_cleaning["srr_id"] + "/" + meta_cleaning["srr_id"] + "_2.fastq"
)
# save it as a csv
meta_cleaning.to_csv("data/metadata_cleanNF.csv", header=False, index=False)
display(meta_cleaning)
```

1.2 `.config` file

With the data made now, we can run the Nextflow pipeline and also use a `.config` file for running the pipeline using the Sun Grid Engine queue system. This is useful to provide the arguments of our pipeline: - threads: argument to control how many CPUs are used for each step - outdir: the path where the output file will be stored - metadata: the path to the input CSV table - conda_env: general Conda environment to run the principal programs - fastp_env: the Conda environment where `fastp` is stored - trimF: the number of bases to trim from the front when it is required

```
#| eval: false

/*
This profile is made to run the the preprocesing pipeline in a cluster related to the profile in this scri
or in local if the local profile is selected
*/
params {
    //main options
    threads = 8
    outdir = "results/nf/clean_data"
    metadata = "data/metadata_cleanNF.csv"
    conda_env= "/export/space3/users/ismadlsh/conda/bio_informatics"
    fastp_env= "fastp"
    //fastp
    trimF=15
}
profiles {
    //configure 2 profiles

    //local running
    local {
        process{
            executor = 'local'
            cpus = 8
            memory = '16 GB'
        }
    }
    //One for sungride engine
    sge{
        //general configuration for all process
        process{
            executor = "sge"
```

```
            penv ="smp"
            cpus = 8
            memory = '10 GB'
            clusterOptions = "-q default -cwd -l h_rt=05:00:00"
        }
    }
}
```

1.3 Run and process

Run it using `nextflow` and the profile `sge`

```
#| eval: false
nextflow run src/clean_data.nf -c src/clean_data.config -profile sge
```

The pipeline has the following processes that are controlled by 2 workflows:

1.3.1 Workflows

To control the execution of all the processes in the pipeline, we have two workflows: the implicit main workflow and the `report_cleaned` workflow in order to clean the data, as well as make the **QC** reports from both:

```
//for making the reports of the cleaning step
workflow report_cleaned {

    take:
        cleaned_fastqs

    main:
        clean_fastqc = fastQC(cleaned_fastqs).fastq_out

        clean_fastqc
            .map { _id, zip_pths -> zip_pths }
            .flatten()
            .collect()
            .set { all_fastqc_clean }

        multiQC( all_fastqc_clean.map{ tuple("clean_data", it) } )
}

//main implicit workflow
workflow {
    //check if the user is providing paired data or not
    meta_ch = channel.fromPath(params.metadata)
        .splitCsv(header: false, sep: ",")
        .map {row ->
            def reads = row.size() > 2 && row[2] ?
            [ file(row[1]), file(row[2]) ] :
            [ file(row[1]) ]
            tuple(row[0], reads)
        }

    //QC proccess
    raw_fastqc=fastQC(meta_ch.map{ _id, _paths -> tuple (_id , _paths, "raw_data")}).fastq_out
    //merge the with the data paths
    combine_ch=meta_ch.join(raw_fastqc)

    //clean the data
```

```
    clean_data=fastp(combine_ch).fastq_clean
    //now we can make the fastqc reports of the clean data
    report_cleaned(clean_data.map{ _id, _paths -> tuple (_id , _paths, "clean_data")})

    //condensate all the fastqc
    raw_fastqc
        .map { _id, zip_pths -> zip_pths}
        .flatten()
        .collect()
        .set { all_fastqc_raw }

    //now multiqc the reports
    multiQC( all_fastqc_raw.map{ tuple("raw_data", it) } )
}
```

So in this workflow, first we read the table in order to obtain the paths of the data and the SRR ID, even if they are paired or unpaired ones. With this first **channel** created, we can move the info beyond the processes:

1.3.2 FastQC

This process gets a tuple that, based on the workflow, comes from a channel. This tuple has not only the parsed data from the table, but also the tag if it is the cleaned data or the raw one, this to control the output directories where the data is going to be saved. This data goes out from the table containing the SRR ID as well as the generated FastQC reports, only the zip ones, that contain all the information that we would require in the next steps.

```
process fastQC {
    //name of the nextflow job
    tag "${srr_id}_QCreps_${type}"
    //save options
    publishDir "${params.outdir}/${type}/FastQC/${srr_id}", mode: 'copy'

    input:
        tuple val(srr_id), path(srr_files),val(type)
    output:
        tuple val(srr_id), path("*_fastqc.zip"), emit: fastq_out
    script:
    """
    fastqc -t ${params.threads} ${srr_files.join(' ')}
    """
}
```

1.3.3 fastp

The next process, naturally, is to clean the data. This is done by the `fastp` process, which takes as input another **tuple** with the SRR ID and the `.zip` results from the `FastQC` process in the way that we presented above. In addition, the `fastq` paths that we parsed in order to have the files for the cleaning process:

```
process fastp {
    //name of the nextflow job
    tag "${srr_id}_fastp"
    //save options
    publishDir "${params.outdir}/clean_data/clean_fastq/${srr_id}", mode: 'copy'

    input:
        tuple val(srr_id), path(srr_files), path(fastqc_res)
    output:
        tuple val(srr_id), path("*_clean.fastq"), emit: fastq_clean
```

```
    script:
    """
    #make this for every zip file
    bash ${projectDir}/run_fastp.sh ${params.trimF} ${params.fastp_env} ${fastqc_res.join(' ')}


    """
}
```

1.3.4 MultiQC

Finally, the pipeline condenses all the QC reports in a MultiQC step at the end of each **quality control** step with all the `.zip` files generated, and for saving the information, also the tag if it is `clean_data` or `raw_data`:

```
process multiQC{
    //name of the nextflow job
    tag "multiQC_${type}"
    //save options
    publishDir "${params.outdir}/${type}/multiQC/", mode: 'copy'

    input:
        tuple val(type), path(fastqc_reps)
    output:
        path("*.html"), emit: multiqc_res
    script:
    """
    multiqc ${fastqc_reps.join(' ')}
    """
}
```

1.3.5 Overview

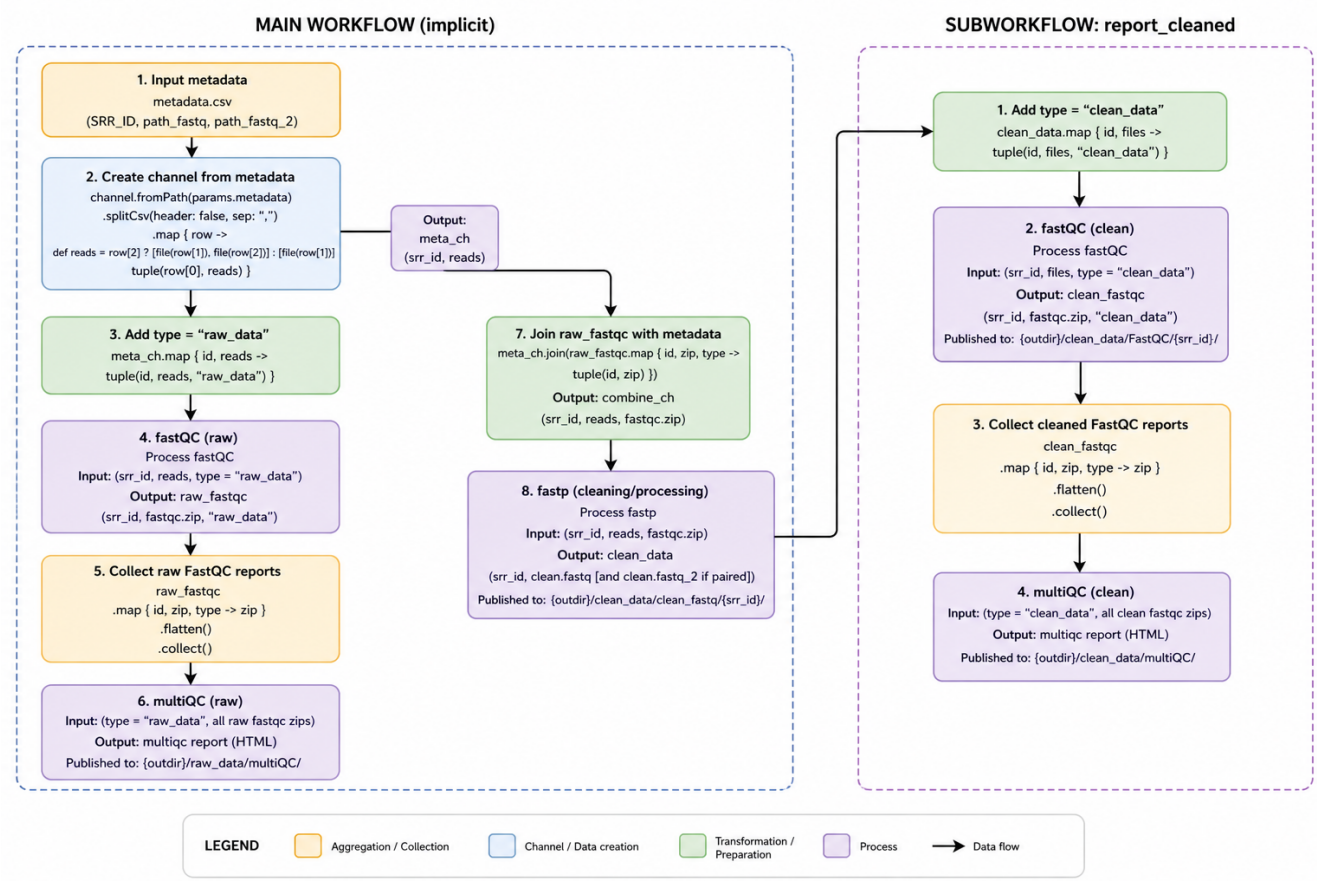


Figure 1: Pipeline Overview

1.4 Results

This generate the following structure:

```
#| eval: false

results/nf_cleaning
   clean_data
       clean_fastq
           SRR9126859
               SRR9126859_1_clean.fastq
               SRR9126859_2_clean.fastq
           SRR9126934
               SRR9126934_1_clean.fastq
               SRR9126934_2_clean.fastq
           SRR9127010
               SRR9127010_1_clean.fastq
               SRR9127010_2_clean.fastq
           SRR9127157
               SRR9127157_1_clean.fastq
               SRR9127157_2_clean.fastq
           SRR9127293
               SRR9127293_1_clean.fastq
               SRR9127293_2_clean.fastq
           SRR9127296
               SRR9127296_1_clean.fastq
               SRR9127296_2_clean.fastq
           SRR9127369
               SRR9127369_1_clean.fastq
               SRR9127369_2_clean.fastq
           SRR9127396
                SRR9127396_1_clean.fastq
                SRR9127396_2_clean.fastq
       FastQC
           SRR9126859
               SRR9126859_1_clean_fastqc.zip
               SRR9126859_2_clean_fastqc.zip
           SRR9126934
               SRR9126934_1_clean_fastqc.zip
               SRR9126934_2_clean_fastqc.zip
           SRR9127010
               SRR9127010_1_clean_fastqc.zip
               SRR9127010_2_clean_fastqc.zip
           SRR9127157
               SRR9127157_1_clean_fastqc.zip
               SRR9127157_2_clean_fastqc.zip
           SRR9127293
               SRR9127293_1_clean_fastqc.zip
               SRR9127293_2_clean_fastqc.zip
           SRR9127296
               SRR9127296_1_clean_fastqc.zip
               SRR9127296_2_clean_fastqc.zip
           SRR9127369
               SRR9127369_1_clean_fastqc.zip
               SRR9127369_2_clean_fastqc.zip
           SRR9127396
                SRR9127396_1_clean_fastqc.zip
```

```
            SRR9127396_2_clean_fastqc.zip
    multiQC
        multiqc_report.html
raw_data
    FastQC
        SRR9126859
            SRR9126859_1_fastqc.zip
            SRR9126859_2_fastqc.zip
        SRR9126934
            SRR9126934_1_fastqc.zip
            SRR9126934_2_fastqc.zip
        SRR9127010
            SRR9127010_1_fastqc.zip
            SRR9127010_2_fastqc.zip
        SRR9127157
            SRR9127157_1_fastqc.zip
            SRR9127157_2_fastqc.zip
        SRR9127293
            SRR9127293_1_fastqc.zip
            SRR9127293_2_fastqc.zip
        SRR9127296
            SRR9127296_1_fastqc.zip
            SRR9127296_2_fastqc.zip
        SRR9127369
            SRR9127369_1_fastqc.zip
            SRR9127369_2_fastqc.zip
        SRR9127396
            SRR9127396_1_fastqc.zip
            SRR9127396_2_fastqc.zip
    multiQC
        multiqc_report.html
```

1.5 Analysis

So now we are able to compare both the MultiQC with the raw data and the MultiQC generated with the pipeline that uses an automatic fastq cleaning process:

1.5.1 Raw data

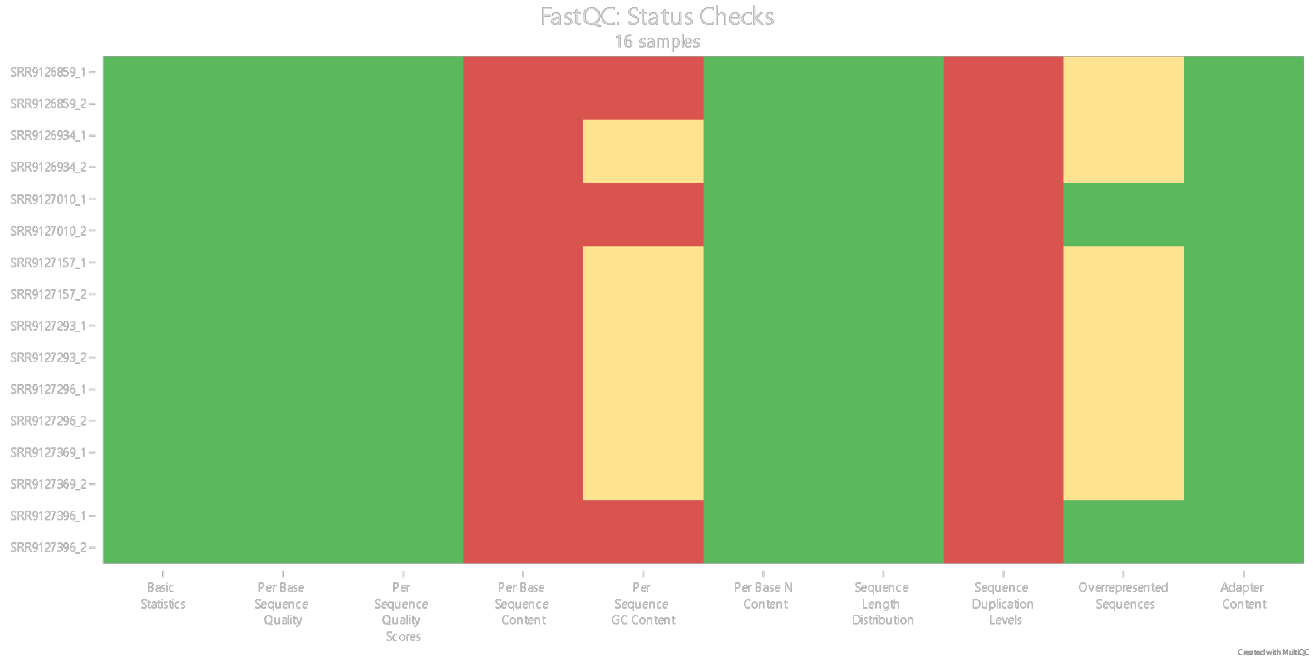


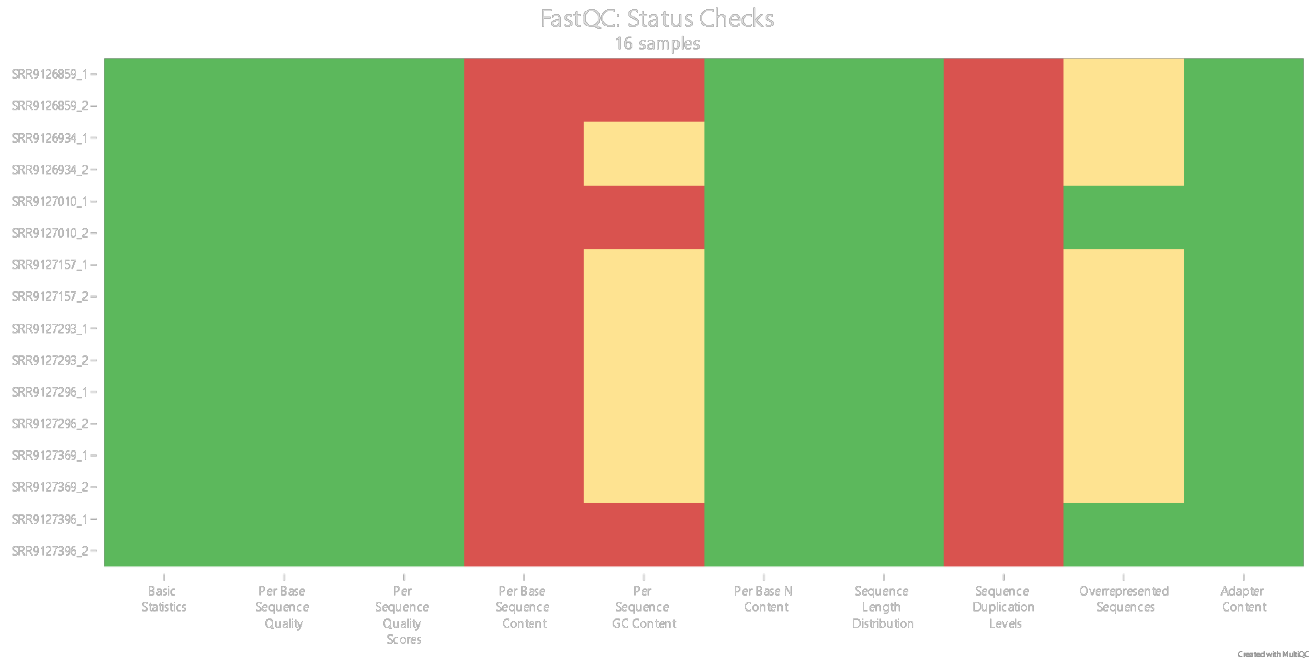
Figure 2: Raw MultiQC

We can see that the general problems of the data are:

- Per base sequence content: this is due to the noise that is intrinsic to the first bases that are present in the sequences, representing a clear bias
- Per sequence GC content: This field is also a problem because it could be an insight into contamination due to the inconsistencies in the GC content
- Sequence Duplication Levels: This is also one of the problems that are more commonly present in fastq data; this should be because of the ribosomal RNA
- Overrepresented sequences: the sequences that are overrepresented are related to the duplication levels due to ribosomal RNA

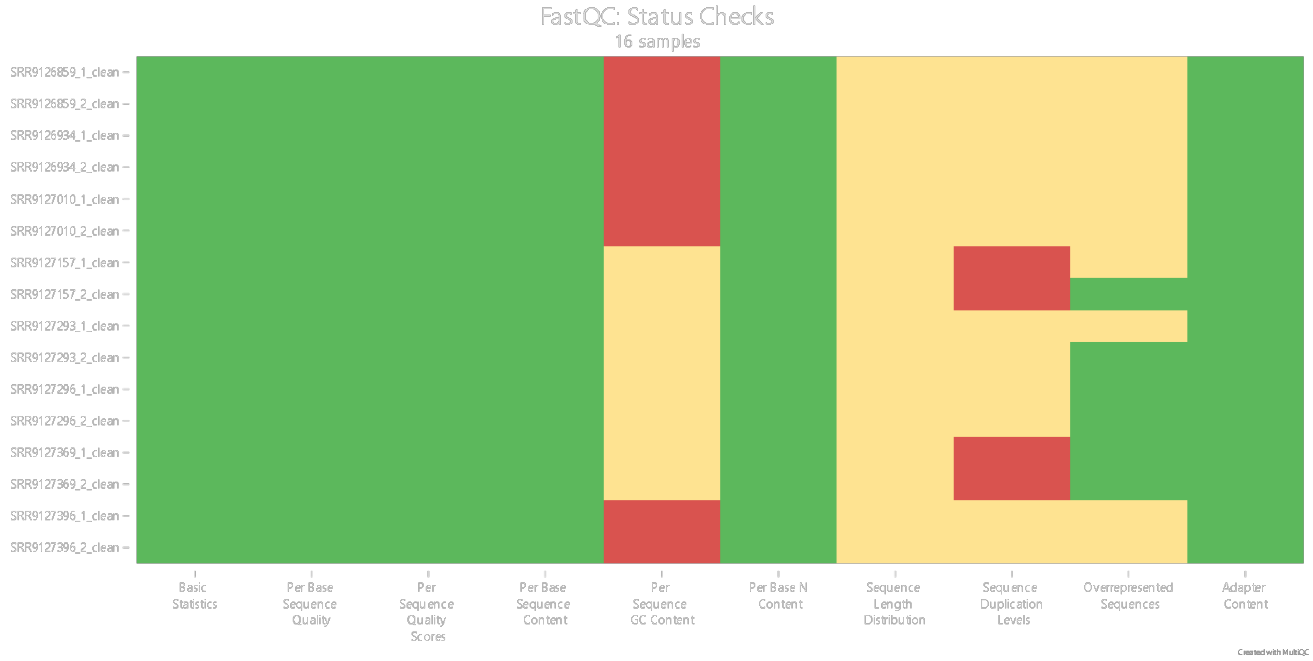
So the pertinent fastp flags were used in order to clean and account for some of those problems.

1.5.2 Original preprocessing



We can see that as we did not use an analysis of every FastQC report in the previous step in a semi-autonomous way, the corrections related to the previous problems are not contundent or even there are no changes.

1.5.3 Clean data



When we performed the cleaning in order with the `FAIL` features present in the `zip` files from the **FastQC** reports, we can see that some problems like the **duplication levels** or the **base sequence content** are fixed or partially fixed. However, this could be a little problem in the **FastQC** as we are cleaning those reads that have low consistency at the beginning of the reads. This is why now not all the libraries have the same length distribution.

Finally, some problems such as the GC bias are not corrected here but should be in next steps using options from other tools like `salmon`.

1.6 Conclusion

While parsing and processing a significant number of samples or libraries from sequencing experiments, it is important to not do them one by one, as the computational tools and frameworks allow parallelization and **sequentialization** of the processes. This is particularly the basis of pipelines and workflow managers like Nextflow. Nextflow is a very useful tool as we can combine dynamic results using **channels**, multiprocessing with their **functions** and **processes**, all controlled by one or more **workflows**. This is particularly useful when extracting statistics from one step in order to be used for cleaning (as is the example of this made pipeline). This combination of the own directories that Nextflow uses and manages in the `work` directory makes it easier to manage the relative and absolute routes in the input step, as Nextflow uses symbolic links to manage all the external files and the output files, in order to not only save the results in a specific path but also to select what outputs of the process we are saving.

1.7 Code

All the code and extended results are disponible in Tabula_Muris_Transcriptomic_analysis git hub repository